# Sales Performance & Revenue Analysis Dashboard

# 1.Objective

TrendMart Retail required a centralized reporting solution to analyze sales performance, profitability, product performance, and regional trends.
The goal was to design and implement an end-to-end Business Intelligence solution using SQL Server, SSIS, and SSRS to support operational and executive-level decision making.

# 2.Business Problem

Retail management lacked visibility into:

- Monthly revenue trends
- Profit margin performance
- Region-wise sales contribution
- Top-performing products
- Customer segmentation revenue
- Year-over-Year growth

Reports were manually prepared in Excel, leading to inconsistencies and delays.
This project delivers an automated, scalable BI reporting solution.

# 3.Technology Stack

| Layer | Tool |
|---|---|
| Database | SQL Server |
| ETL | SQL Server Integration Services (SSIS) |
| Reporting | SQL Server Reporting Services (SSRS) |
| Data Source | CSV files (Simulated Retail Data) |

# 4.Data Architecture

## 4.1 Data Model Design

A **Star Schema** was implemented to optimize reporting performance and simplify analytical queries.

## 4.2 Fact Table

FactSales

### 4.3 Dimension Tables

- DimDate
- DimCustomer
- DimProduct
- DimRegion

### 4.4 Why Star Schema?

- Faster aggregations
- Simplified joins
- BI-friendly design
- Supports scalable reporting

FactSales contains transactional data, while dimension tables provide descriptive attributes.

## 5.Database Schema Design

### 5.1 FactSales

- SalesID (Primary Key)
- OrderNumber
- DateKey (FK)
- CustomerID (FK)
- ProductID (FK)
- RegionID (FK)
- Quantity
- UnitPrice
- TotalRevenue
- TotalCost
- Profit

### 5.2 DimCustomer

- CustomerID
- CustomerName
- Segment
- City
- State

### 5.3 DimProduct

- ProductID
- ProductName
- Category
- UnitCost

### 5.4 DimRegion

- RegionID
- RegionName
- Country

★ Sales-Performance-Dashboard/Database/Dimension Tables.sql

## 6.ETL Process – SSIS Implementation

### 6.1 Data Source

CSV files were used to simulate raw retail transaction data.

★ Sales-Performance-Dashboard/Database/Custimer.csv
★ Sales-Performance-Dashboard/Database/Products.csv
★ Sales-Performance-Dashboard/Database/Regions.csv
★ Sales-Performance-Dashboard/Database/Sales_Transactions.csv

### 6.2 ETL Workflow

Control Flow Execution Order:

1. Load DimCustomer
2. Load DimProduct
3. Load DimRegion
4. Load FactSales

Dimensions are loaded first to maintain referential integrity.

★ Sales-Performance-Dashboard/SSIS/Package.dtsx

### 6.3 Data Validation Performed

✓ Null value checks

✓ Duplicate removal

✓ Data type validation

✓ Standardization (Trim, Uppercase)

✓ Cost rounding

✓ Foreign key lookups

Invalid rows were redirected for error handling.

### 6.4 KPI Calculations in ETL

Calculated inside Data Flow using Derived Columns:

- TotalRevenue = Quantity × UnitPrice
- TotalCost = Quantity × UnitCost
- Profit = TotalRevenue − TotalCost

This ensures fact table stores pre-calculated measures for performance optimization.

## 7.Reporting Layer – SSRS

Two reports were developed:

### 7.1 Sales Performance Dashboard (Operational Report)

Purpose:
Provide sales managers with interactive performance tracking.
Includes:

- KPI Strip
    - Total Revenue
    - Total Profit
    - Profit Margin %
    - Total Orders
    - Avg Order Value
- Monthly Revenue Trend (Line Chart)
- Top 5 Products by Revenue (Bar Chart)
- Region Revenue vs Profit (Clustered Column Chart)
- Revenue by Customer Segment (Pie Chart)

This report supports daily and monthly performance review.

### 7.2 Executive KPI Summary (Board-Level Report)

Purpose:
Provide high-level strategic overview for leadership.
Includes:

- Total Revenue
- Total Profit
- Profit Margin %
- YoY Growth %
- Top Performing Region
- Top Product
- Business Insight Section

Conditional formatting applied for growth indicators.

## 8.Key KPIs Delivered

- Total Revenue
- Total Profit
- Profit Margin %
- Year-over-Year Growth %
- Average Order Value
- Revenue by Region
- Product Ranking
- Customer Segment Contribution

## 9.Analytical SQL Highlights

Implemented:

- Aggregations (SUM, COUNT, AVG)
- GROUP BY with ranking
- Window Functions (RANK, DENSE_RANK)
- Year-over-Year calculations
- Contribution percentage analysis

★ Sales-Performance-Dashboard/Database/Analytical Queries.sql

## 10.Business Impact

The solution enables:

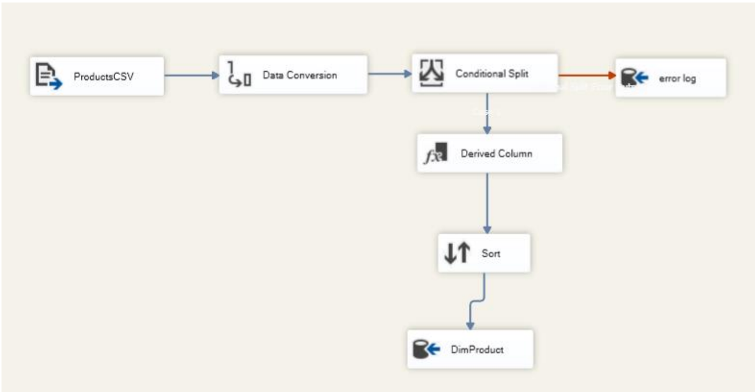
- Real-time performance monitoring
- Executive-level profitability analysis
- Identification of high-performing products
- Regional growth tracking
- Data-driven strategic decisions

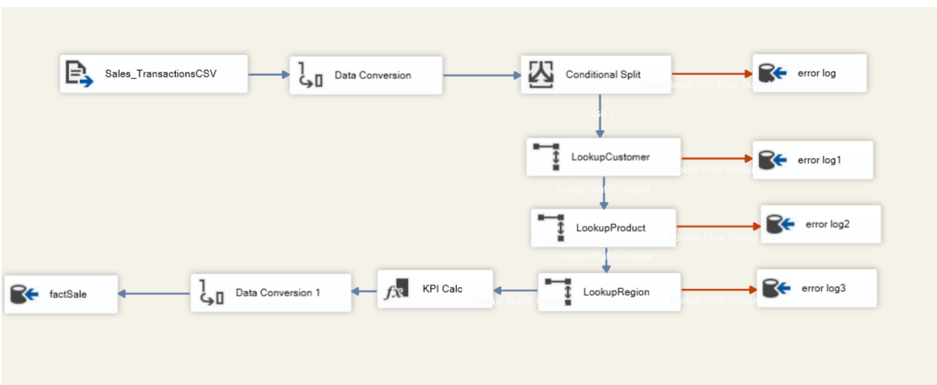
The automated ETL pipeline reduces manual reporting effort and improves data consistency.

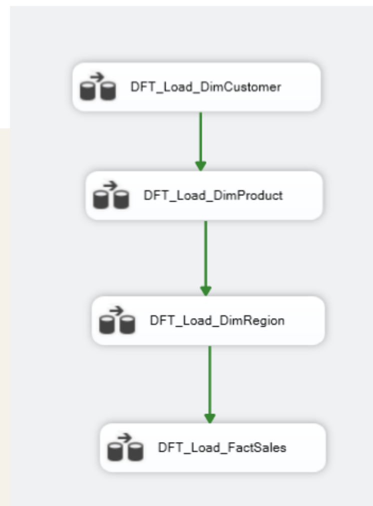
## 11.Future Enhancements

- Add incremental data loading
- Implement Slowly Changing Dimensions
- Add Drillthrough reports
- Integrate Power BI version
- Deploy to Report Server

## 12.Screenshots



YearParam 2026 View Report

1 of 1 100% Find | Next

Executive KPI Summary 2026

| Total Revenue | Total Profit | Profit Margin | YoY Growth |
|---|---|---|---|
| $2,825.00 | $965.00 | 34.16% | 24.00% |

**Top Performing Region:**

MIDWEST

**Top Product**

Smartphone X

**Business Insights**

Revenue growth driven primarily by Mid West.
Profit margin improved due to higher sales of premium products.
Q3 shows seasonal spike in revenue.
Corporate segment remains stable contributor.

YearParam 2026 View Report

1 of 1 75% Find | Next

**Sales Performance Dashboard** Year : 2026

| Total Revenue | Total Profit | Profit Margin % | Total Orders | Avg Order Value |
|---|---|---|---|---|
| $2,825.00 | $965.00 | $34.16 | 2 | $1,412.50 |